

Marking Scheme

2026 · Paper Paper 1 · 105 marks

General Rules

- M / A / PROOF mark types
- **ft.** : M marks always follow through. A marks follow through ONLY when isFollowThrough = true.
- Deduction cap : A(1)=1 · A(2)=1 · B=0 (u, pp)
- No deduction in any step scoring 0.
- A correct answer merits all marks unless a method is specified.
- Different notation is NOT penalized. · Benefit of doubt
- All fractional answers must be simplified.

Section A(1) (35 marks)

1. Laws of Indices [B-]

a (3 marks)

M1 $(x^3y^{-2})^{-2} = x^{-6}y^4$

M1 $(x^{-1}y)^3 = x^{-3}y^3; \frac{x^{-6}y^4}{x^{-3}y^3} = x^{-3}y$

A1 $\frac{y}{x^3}$

2. Linear Equations [B-]

a (3 marks)

M1 $x - y = 96$ and $5y = x$

M1 $5y - y = 96 \Rightarrow 4y = 96$

A1 $y = 24$

3. Algebraic Fractions [B-]

a (3 marks)

M1 $2x - 6 = 2(x - 3)$, so the L.C.M. of the denominators is $2(x - 3)$

M1 $= \frac{4}{2(x - 3)} + \frac{5}{2(x - 3)} = \frac{9}{2(x - 3)}$

A1 $\frac{9}{2x - 6}$

4. Factorization [B]

a (2 marks)

M1 $4m^2 - 12m + 9 = (2m)^2 - 2(2m)(3) + 3^2$

A1 $= (2m - 3)^2$

b (2 marks)

M1 $= (2m - 3)^2 - (5n)^2$ **ft.**
1M for using (a) and difference of two squares

A1 $= (2m - 5n - 3)(2m + 5n - 3)$ **ft.**

5. Percentages [B]

a (4 marks)

M1 $Cost = \frac{160}{20} = \text{\textcolor{red}{800}}$

M1 $Selling price = 800 + 160 = \text{\textcolor{red}{960}}$

M1 $Let the marked price be \text{\textcolor{red}{M}}; M(1 - 25$

A1 $M = \text{\textcolor{red}{1280}}$
u-1 if the dollar sign is missing

6. Compound Inequalities [B]

a (3 marks)

M1 $-3x + 3 \geq x + 19 \Rightarrow -4x \geq 16 \Rightarrow x \leq -4$

M1 $5x < -10 \Rightarrow x < -2$

A1 *Taking the union* : $x < -2$

b (1 mark)

A1 -3 ft.

7. Transformation and Coordinates [B]

a (2 marks)

A1 $U' = (6, 8)$

A1 $V' = (-5, -9)$

b (2 marks)

M1 $\text{Slope} = \frac{-9 - 8}{-5 - 6}$ ft.

A1 $= \frac{17}{11}$ ft.

8. Congruent Triangles [B+]

a (2 marks)

M1 $AB = CD(\text{given}), \angle ABD = \angle CDB(\text{given}), BD = DB(\text{common side})$

PROOF $\therefore \triangle ABD \cong \triangle CDB(SAS)$

b (3 marks)

M1 $\angle CBD = \angle ADB = 27^\circ(\text{corr. } \angle\text{s}, \cong \triangle\text{s})$ ft.

M1 $\text{In } \triangle BCD : \angle BDC = 180^\circ - 94^\circ - 27^\circ = 59^\circ(\angle\text{sum of } \triangle)$

A1 $\angle ADC = \angle ADB + \angle BDC = 27^\circ + 59^\circ = 86^\circ$ ft.

9. Frequency Distribution [B]

a (3 marks)

A1 $x = 22 - 6 = 16$

M1 $\text{Mean} = \frac{6(12) + 16(17) + 14(22) + 4(27)}{40} = \frac{760}{40}$ ft.

A1 $= 19\text{hours}$ ft.
u-1 if unit missing

b (2 marks)

M1 $P = \frac{22}{40}$ ft.

A1 $= \frac{11}{20}$ ft.

Section A(2) (35 marks)

10. Partial Variation and Quadratic Equations [A-]

a (3 marks)

M1 *Let* $f(x) = hx^2 + mx$, where $h \neq 0$ and $m \neq 0$

M1 $9h + 3m = 45$ and $4h - 2m = 10$

A1 $h = 4, m = 3 \Rightarrow f(x) = 4x^2 + 3x$

b (1 mark)

A1 $6f(x) = 6x(4x + 3) = 0 \Rightarrow x = 0$ or $x = -\frac{3}{4}$ ft.

c (2 marks)

M1 $4x^2 + 3x - k = 0; \Delta = 3^2 - 4(4)(-k) = 9 + 16k > 0$ ft.

A1 $k > -\frac{9}{16}$ ft.

11. Stem-and-Leaf Diagram and Dispersion [A-]

a (3 marks)

M1 *There are 16 data; median* $= \frac{8\text{th} + 9\text{th}}{2} = \frac{29 + (30 + a)}{2} = 31$

A1 $a = 3$

A1 $Q_1 = \frac{20 + 23}{2} = 21.5, Q_3 = \frac{41 + (40 + b)}{2}; Q_3 - 21.5 = 21 \Rightarrow b = 4$

b (2 marks)

M1 *The mode is 23, which occurs 3 times; every other age occurs at most once* ft.

A1 *After one player leaves, 23 still occurs at least twice, so the mode remains 23 : NO change*

c (2 marks)

M1 *Range* $= 48 - 15 = 33$; *it decreases only if the player aged 15 or the player aged 48 leaves* ft.

A1 *Removing 15 gives standard deviation* ≈ 10.8 , *removing 48 gives* ≈ 10.8 ; *the greatest is 10.8 (corr. to 3 sig. fig.)*
ft.
r.t. 10.8

12. Circles and Coordinate Geometry [A-]

a (3 marks)

M1 $G = \left(\frac{80}{2}, \frac{60}{2} \right) = (40, 30)$

M1 $GH = \sqrt{(45 - 40)^2 + (42 - 30)^2} = \sqrt{25 + 144}$ ft.

A1 $GH = 13$ ft.

b (4 marks)

M1 $GP = \text{radius} = \sqrt{40^2 + 30^2 + 4556} = 84$ (a constant)

A1 $\text{Area} = \frac{1}{2}(GH)(GP) \sin \angle HGP$ is greatest when $\sin \angle HGP = 1$, i. e. $GH \perp GP$

M1 $HP = \sqrt{13^2 + 84^2} = \sqrt{7225} = 85$ ft.

A1 $\text{Perimeter} = 13 + 84 + 85 = 182$ ft.

13. Mensuration and Similarity [A-]

a (3 marks)

M1 *Ratio of volumes* $= 27 : 64 \Rightarrow \text{ratio of radii} = 3 : 4$

M1 *Radius of the smaller sphere* $= \frac{3}{4}(12) = 9 \text{ cm}$

A1 *Surface area* $= 4\pi(9)^2 = 324\pi \text{ cm}^2$ ft.

b (4 marks)

M1 *Total volume* $= \frac{4}{3}\pi(9)^3 + \frac{4}{3}\pi(12)^3 = 972\pi + 2304\pi = 3276\pi \text{ cm}^3$

M1 *Volume of A* $= \frac{1}{3}\pi(9)^2(12) = 324\pi \text{ cm}^3$

M1 *Volume of B* $= 3276\pi - 324\pi = 2952\pi \text{ cm}^3$ ft.

A1 *If A and B were similar, the ratio of radii* $= 18 : 9 = 2 : 1$ *would give volume of B* $= 8(324\pi) = 2592\pi \neq 2952\pi$, *so the claim is NOT correct*
ft.

14. Polynomials [A]

a (3 marks)

M1 $p(1) = 3 \Rightarrow 3 + a + b - 24 = 3 \Rightarrow a + b = 24$

M1 $p(-2) = 0 \Rightarrow -24 + 4a - 2b - 24 = 0 \Rightarrow 2a - b = 24$

A1 $a = 16, b = 8$

b (2 marks)

M1 $p(2) = 3(8) + 16(4) + 8(2) - 24 = 24 + 64 + 16 - 24 = 80$ ft.

A1 $p(2) \neq 0$, *so by the factor theorem* $x - 2$ *is NOT a factor of* $p(x)$ ft.

c (3 marks)

M1 $p(x) = (x + 2)(3x^2 + 10x - 12)$ ft.

M1 *For* $3x^2 + 10x - 12 = 0 : \Delta = 10^2 - 4(3)(-12) = 244 > 0$ *and 244 is not a perfect square*

A1 *So this quadratic has two irrational roots while* $x - 2$ *is rational; the claim is AGREED*
ft.

15. Probability [A-]

a (2 marks)

$$\begin{aligned} \text{M1} \quad P &= \frac{C_2^7 \times C_2^9}{C_4^{16}} = \frac{21 \times 36}{1820} \\ \text{A1} \quad &= \frac{27}{65} \end{aligned}$$

b (2 marks)

$$\begin{aligned} \text{M1} \quad P &= 1 - \frac{27}{65} \quad \text{ft.} \\ &\text{1M for using the complementary event / 使用互補事件} \\ \text{A1} \quad &= \frac{38}{65} \quad \text{ft.} \end{aligned}$$

16. Quadratic Functions [A-]

a (2 marks)

$$\begin{aligned} \text{M1} \quad g(x) &= 2(x^2 + 6kx) + 20k^2 + 6 = 2(x + 3k)^2 - 18k^2 + 20k^2 + 6 \\ \text{A1} \quad g(x) &= 2(x + 3k)^2 + 2k^2 + 6; \text{vertex} = (-3k, 2k^2 + 6) \end{aligned}$$

b (3 marks)

$$\begin{aligned} \text{M1} \quad 3g(-x) &= 6x^2 - 36kx + 60k^2 + 18 = 6(x - 3k)^2 + 6k^2 + 18 \quad \text{ft.} \\ \text{A1} \quad B &= (3k, 6k^2 + 18) \quad \text{ft.} \\ \text{A1} \quad M &= \left(\frac{-3k + 3k}{2}, \frac{(2k^2 + 6) + (6k^2 + 18)}{2} \right) = (0, 4k^2 + 12) \quad \text{ft.} \end{aligned}$$

17. Roots of Equations and Sequences [A]

a (3 marks)

$$\begin{aligned} \text{M1} \quad \alpha + \beta &= -c \text{ and } \alpha\beta = -12 \\ \text{M1} \quad \alpha^2 + \beta^2 &= (\alpha + \beta)^2 - 2\alpha\beta \\ \text{A1} \quad &= c^2 + 24 \end{aligned}$$

b (4 marks)

$$\begin{aligned} \text{M1} \quad \text{Common difference} &= (c^2 + 24) - c^2 = 24 \quad \text{ft.} \\ \text{A1} \quad 3\text{rd term} = c^2 + 48 = 76 \Rightarrow c^2 = 28; \text{the 1st term is } 28 \\ \text{M1} \quad S(n) &= \frac{n}{2}[2(28) + (n-1)(24)] = 12n^2 + 16n > 3 \times 10^5 \quad \text{ft.} \\ \text{A1} \quad S(499) &= 2995996 < 3 \times 10^5 \cdot 10 \text{ and } S(500) = 3008000 > 3 \times 10^5 \cdot 10; \text{least } n = 500 \quad \text{ft.} \end{aligned}$$

18. Trigonometry in 3-D [A]

a (3 marks)

$$\begin{aligned} \text{M1} \quad \text{By the cosine formula: } BC^2 &= 24^2 + 20^2 - 2(24)(20)\cos 105^\circ \\ \text{M1} \quad BC^2 &= 976 + 248.47 = 1224.47 \\ \text{A1} \quad BC &\approx 35.0 \text{ cm} \\ &\text{u-1 if unit cm missing} \\ &\text{r.t. 35.0} \end{aligned}$$

b (4 marks)

$$\begin{aligned} \text{M1} \quad \text{Vertical height of C above the ground} &= AC \sin 60^\circ = 20 \sin 60^\circ \approx 17.32 \text{ cm} \\ \text{M1} \quad \text{Blies on the ground, so the height of N (mid-point of BC)} &= \frac{17.32}{2} \approx 8.66 \text{ cm} \quad \text{ft.} \\ \text{M1} \quad \text{The horizontal distance from A to N is found to be} &\approx 10.34 \text{ cm} \quad \text{ft.} \\ \text{A1} \quad \tan \theta = \frac{8.66}{10.34} \Rightarrow \theta \approx 40.0^\circ > 35^\circ, \text{ so the claim is correct} &\quad \text{ft.} \\ &\text{r.t. 40.0} \end{aligned}$$

19. Circles and Tangents [A+]

a (2 marks)

M1 $Slope of AG = \frac{120 - 40}{90 - 30} = \frac{4}{3}$

A1 $y - 40 = \frac{4}{3}(x - 30) \Rightarrow 4x - 3y = 0$

b (3 marks)

M1 $AG = \sqrt{60^2 + 80^2} = 100$

M1 $PQ \perp AG; \text{ let } M \text{ be the intersection. In right-angled } \triangle GPA, GM = \frac{GP^2}{GA} = \frac{50^2}{100} = 25$

A1 $M = (30, 40) + 25 \left(\frac{3}{5}, \frac{4}{5} \right) = (45, 60)$ ft.

c (4 marks)

M1 $AP = \sqrt{100^2 - 50^2} = 50\sqrt{3}$ and $PQ = 2 \times \frac{GP \cdot AP}{GA} = 50\sqrt{3}$ ft.

M1 $AP = AQ = PQ$, so $\triangle APQ$ is equilateral and its in-centre is the centroid, lying on AM

M1 $AM = \sqrt{45^2 + 60^2} = 75$; in-centre $= A + \frac{2}{3}(M - A) = (60, 80)$ and in-radius $= \frac{75}{3} = 25$
ft.

A1 $(x - 60)^2 + (y - 80)^2 = 625$, i.e. $x^2 + y^2 - 120x - 160y + 9375 = 0$ ft.

d (3 marks)

M1 For the equilateral $\triangle APQ$ the centroid, in-centre and circumcentre coincide at $(60, 80)$
ft.

M1 Circumradius $= \frac{2}{3}(AM) = 50$, in-radius $= 25$ ft.

A1 Ratio of areas $= \pi(25)^2 : \pi(50)^2 = 1 : 4$, so the claim is AGREED ft.